Executive summary (added after the run; numbers below are copied from the sections that follow)

Verdict: NO-GO. The six-phase roadmap was implemented end to end (data pipeline, engine, baseline reproduction, protocol, 17 ideas / 127 configurations, risk layer, locked test).

• **Baseline reproduced** within the roadmap's tolerance: 17.0% CAGR / -13.4% MaxDD / 0.90 Sharpe over Apr 2017-Oct 2026 (paper: 17.1% / -12.8% / 0.94) and recent-regime Sharpe 0.57 (paper 0.56). It only reproduces with floor costs + 6% interest on collateral + Sharpe on excess return, which is the yardstick used throughout.

• **Phase 3**: of 17 ideas only the VIX-percentile filter (stand aside when prior-day VIX 252-day percentile > 80) passed train and validation. Shorts, daily-trend agreement, chop filters, intraday-only, opening range, gaps, RSI mean reversion, OI and US-overnight ideas all failed train or validation. Gate G3 (2-3 ideas from 2 families) was **not met**.

• **Frozen spec**: baseline Supertrend + VIX filter (p=80) + stop/risk sizing k=2.5 (0.75% of capital per trade); vol targeting, overnight cap, drawdown brake and daily loss limit did not earn a place.

• **Locked test (Jan 2024-Oct 2026)**: strategy 5.5% CAGR / -8.7% MaxDD / **-0.09 Sharpe** vs baseline 10.4% / -6.6% / **0.46** (floor costs); with realistic costs 0.09 vs 0.62. Buy-and-hold futures: 2.8% CAGR, -14.6% MaxDD, Sharpe -0.20 (synthetic). The 95% bootstrap CI for the Sharpe difference vs baseline is [-1.45, +0.04]. Pass criteria failed on 6 of 8 lines.

• **What did hold**: no lookahead (truncation, shuffled-future and extra-delay tests), per-trade loss at the stop <= 0.75% of capital (gaps through the stop reach 1.9%), no calendar year below -3.3%, drawdown on train/validation far better than the baseline's.

• **Caveats that matter**: the synthetic futures series is built from spot, whose 09:15 print differs from real futures (section 8: synthetic nights earn more and days less than real futures); a real-futures-anchored path (section 8b) gives the same ranking. Test-window Sharpe has ~0.6 standard error, PBO 49%, 127 configurations tried.

Final report: Nifty 50 futures strategy research roadmap (generated 2026-11-03)

Frozen spec (`results/rm/FROZEN_SPEC.json`, sha256 `196112ad5cbf98ae...`): Supertrend(14,3) Wilder ATR 15-min spot, long-only, stand aside when prior-day India VIX 252d percentile > 80 (B2h); stop k=2.5 x ATR(15-min), lots = min(full, floor(0.0075 x Rs2cr / (k x ATR x 65))); no vol targeting / overnight cap / drawdown brake / daily loss limit (each failed to earn its place on train). Dataset sha256 `b6ca337fe078228d...`.

1. Results by window (floor costs = mandate yardstick; total return incl. 6% interest on collateral, Sharpe on excess return)

Mandate floor costs (0.025%/side, 0.05%/roll)

	CAGR	MaxDD	Sharpe	Sortino	Calmar	Vol	TradesPerYr	WinRate	AvgWin/AvgLoss	TimeInMkt	CostDrag %/yr	LongLeg CAGR	ShortLeg CAGR
Train 2017-04..2021-12 — strategy	19.3%	-6.2%	1.68	2.95	3.11	7.1%	67.24	43.1%	2.19	40.7%	3.44	11.4%	0.0%
Train 2017-04..2021-12 — baseline	22.8%	-13.4%	1.15	1.84	1.71	13.3%	87.22	43.3%	1.96	56.7%	4.65	15.0%	0.0%
Validate 2022-01..2023-12 — strategy	10.7%	-5.5%	0.60	0.93	1.94	7.2%	88.61	37.6%	2.06	41.8%	4.44	5.2%	0.0%
Validate 2022-01..2023-12 — baseline	13.3%	-7.7%	0.73	1.13	1.74	9.2%	100.39	41.3%	1.78	54.0%	5.20	7.7%	0.0%
TEST 2024-01..2026-10 (locked, opened once) — strategy	5.5%	-8.7%	-0.09	-0.15	0.64	5.9%	65.69	31.1%	2.09	33.2%	3.36	-1.0%	0.0%

	CAGR	MaxDD	Sharpe	Sortino	Calmar	Vol	TradesPerYr	WinRate	AvgWin/AvgLoss	TimeInMkt	CostDrag%/yr	LongLegCAGR	ShortLegCAGR
TEST 2024-01..2026-10 (locked, opened once) — baseline	10.4%	-6.6%	0.46	0.72	1.58	9.2%	86.15	36.7%	1.99	52.9%	4.46	3.8%	0.0%
Recent 2022-01..2026-10 — strategy	7.6%	-8.7%	0.23	0.35	0.88	6.5%	75.14	34.3%	2.07	36.7%	3.81	1.6%	0.0%
Recent 2022-01..2026-10 — baseline	11.6%	-7.7%	0.57	0.89	1.52	9.2%	92.02	38.8%	1.89	53.4%	4.77	5.4%	0.0%
Full 2017-04..2026-10 — strategy	13.3%	-8.7%	0.98	1.62	1.53	6.8%	71.22	38.4%	2.17	38.7%	3.63	6.5%	0.0%
Full 2017-04..2026-10 — baseline	17.0%	-13.4%	0.90	1.42	1.27	11.4%	89.64	41.0%	1.95	55.0%	4.71	10.1%	0.0%

Realistic costs (dated STT, brokerage, exchange, SEBI, stamp, GST, 1 tick; roll = 2 legs + spread)

	CAGR	MaxDD	Sharpe	Sortino	Calmar	Vol	TradesPerYr	WinRate	AvgWin/AvgLoss	TimeInMkt	CostDrag%/yr	LongLegCAGR	ShortLegCAGR
Train 2017-04..2021-12 — strategy	21.8%	-5.9%	1.98	3.58	3.71	7.0%	67.24	44.1%	2.33	40.7%	1.40	13.5%	0.0%
Train 2017-04..2021-12 — baseline	26.2%	-12.8%	1.36	2.20	2.05	13.3%	87.22	43.8%	2.08	56.7%	1.90	17.7%	0.0%
Validate 2022-01..2023-12 — strategy	13.6%	-4.0%	0.98	1.54	3.44	7.1%	88.61	40.5%	2.04	41.8%	1.86	7.8%	0.0%
Validate 2022-01..2023-12 — baseline	16.8%	-5.7%	1.07	1.68	2.93	9.2%	100.39	44.4%	1.73	54.0%	2.18	10.7%	0.0%
TEST 2024-01..2026-10 (locked, opened once) — strategy	6.7%	-7.6%	0.09	0.14	0.87	5.9%	65.69	31.1%	2.22	33.2%	2.30	0.1%	0.0%
TEST 2024-01..2026-10 (locked, opened once) — baseline	12.0%	-6.3%	0.62	0.96	1.91	9.2%	86.15	37.1%	2.06	52.9%	3.03	5.2%	0.0%
Recent 2022-01..2026-10 — strategy	9.5%	-7.6%	0.49	0.78	1.24	6.4%	75.14	35.7%	2.12	36.7%	2.12	3.3%	0.0%
Recent 2022-01..2026-10 — baseline	14.0%	-6.3%	0.80	1.26	2.21	9.2%	92.02	40.4%	1.90	53.4%	2.68	7.5%	0.0%
Full 2017-04..2026-10 — strategy	15.4%	-7.6%	1.26	2.13	2.03	6.7%	71.22	39.6%	2.26	38.7%	1.76	8.3%	0.0%
Full 2017-04..2026-10 — baseline	19.9%	-12.8%	1.11	1.78	1.55	11.4%	89.64	42.0%	2.01	55.0%	2.30	12.5%	0.0%

Floor costs, WITHOUT interest on idle collateral (pure futures P&L; long pays the basis)

	CAGR	MaxDD	Sharpe	Sortino	Calmar	Vol	TradesPerYr	WinRate	AvgWin/AvgLoss	TimeInMkt	CostDrag%/yr	LongLegCAGR	ShortLegCAGR
Train 2017-04..2021-12 — strategy	12.3%	-7.0%	1.68	2.95	1.75	7.0%	67.24	43.1%	2.19	40.7%	3.44	11.4%	0.0%
Train 2017-04..2021-12 — baseline	15.5%	-14.7%	1.15	1.84	1.06	13.3%	87.22	43.3%	1.96	56.7%	4.65	15.0%	0.0%
Validate 2022-01..2023-12 — strategy	4.2%	-9.3%	0.60	0.93	0.45	7.2%	88.61	37.6%	2.06	41.8%	4.44	5.2%	0.0%
Validate 2022-01..2023-12 — baseline	6.6%	-11.0%	0.73	1.13	0.60	9.2%	100.39	41.3%	1.78	54.0%	5.20	7.7%	0.0%
TEST 2024-01..2026-10 (locked, opened once) — strategy	-0.7%	-13.5%	-0.09	-0.15	-0.05	5.9%	65.69	31.1%	2.09	33.2%	3.36	-1.0%	0.0%
TEST 2024-01..2026-10 (locked, opened once) — baseline	3.9%	-8.2%	0.46	0.72	0.48	9.2%	86.15	36.7%	1.99	52.9%	4.46	3.8%	0.0%
Recent 2022-01..2026-10 — strategy	1.3%	-13.5%	0.23	0.35	0.09	6.4%	75.14	34.3%	2.07	36.7%	3.81	1.6%	0.0%

	CAGR	MaxDD	Sharpe	Sortino	Calmar	Vol	TradesPerYr	WinRate	AvgWin/AvgLoss	TimeInMkt	CostDrag %/yr	LongLeg CAGR	ShortLeg CAGR
Recent 2022-01..2026-10 — baseline	5.0%	-11.0%	0.57	0.89	0.45	9.2%	92.02	38.8%	1.89	53.4%	4.77	5.4%	0.0%
Full 2017-04..2026-10 — strategy	6.6%	-13.5%	0.98	1.62	0.49	6.8%	71.22	38.4%	2.17	38.7%	3.63	6.5%	0.0%
Full 2017-04..2026-10 — baseline	10.1%	-14.7%	0.90	1.42	0.69	11.4%	89.64	41.0%	1.95	55.0%	4.71	10.1%	0.0%

2. Calendar years (total return, floor costs unless stated; 2026 = Jan-Oct)

	strategy	baseline	strategy (real cost)	baseline (real cost)
2017	8.3%	5.5%	10.0%	7.8%
2018	10.0%	10.5%	11.7%	13.6%
2019	13.6%	14.9%	16.3%	18.1%
2020	27.8%	39.7%	30.2%	43.6%
2021	31.7%	39.1%	34.7%	42.5%
2022	11.1%	10.0%	13.8%	13.6%
2023	9.8%	16.0%	12.8%	19.2%
2024	-3.3%	7.7%	-1.7%	10.3%
2025	14.7%	14.7%	16.4%	16.6%
2026	4.7%	6.8%	4.6%	6.6%

3. Rolling walk-forward (3y train, 6m OOS, 6m step; parameters re-picked per fold by the plateau rule; OOS stitched; run once)

	CAGR	MaxDD	Sharpe	Sortino	Calmar	Vol	TradesPerYr	WinRate	AvgWin/AvgLoss	TimeInMkt	CostDrag %/yr	LongLeg CAGR	ShortLeg CAGR
strategy WF	14.5%	-6.2%	1.12	1.87	2.34	6.9%	nan	na	nan	na	nan	na	na
baseline WF	20.8%	-7.7%	1.29	2.12	2.72	10.3%	nan	na	nan	na	nan	na	na

	train	p	k	train_sharpe
2020-04-01..2020-09-30	2017-04-01..2020-03-31	80	2.50	0.73
2020-10-01..2021-03-31	2017-10-01..2020-09-30	80	2.50	1.16
2021-04-01..2021-09-30	2018-04-01..2021-03-31	80	2.50	1.67
2021-10-01..2022-03-31	2018-10-01..2021-09-30	80	2.50	1.98
2022-04-01..2022-09-30	2019-04-01..2022-03-31	80	2.50	1.99
2022-10-01..2023-03-31	2019-10-01..2022-09-30	80	2.50	2.02
2023-04-01..2023-09-30	2020-04-01..2023-03-31	80	2.50	1.77
2023-10-01..2024-03-31	2020-10-01..2023-09-30	80	2.50	1.51
2024-04-01..2024-09-30	2021-04-01..2024-03-31	70	2.50	1.05

	train	p	k	train_sharpe
2024-10-01..2025-03-31	2021-10-01..2024-09-30	70	2.50	0.16
2025-04-01..2025-09-30	2022-04-01..2025-03-31	70	2.50	-0.16
2025-10-01..2026-03-31	2022-10-01..2025-09-30	70	2.50	-0.10
2026-04-01..2026-09-30	2023-04-01..2026-03-31	70	2.50	0.63
2026-10-01..2026-10-31	2023-10-01..2026-09-30	70	2.50	0.31

4. Stress tests on the test window (reported, never used to pick parameters)

	CAGR	MaxDD	Sharpe	Sortino	Calmar	Vol	TradesPerYr	WinRate	AvgWin/AvgLoss	TimeInMkt	CostDrag %/yr	LongLeg CAGR	ShortLeg CAGR
strategy, floor	5.5%	-8.7%	-0.09	-0.15	0.64	5.9%	65.69	31.1%	2.09	33.2%	3.36	-1.0%	0.0%
baseline, floor	10.4%	-6.6%	0.46	0.72	1.58	9.2%	86.15	36.7%	1.99	52.9%	4.46	3.8%	0.0%
strategy, costs x2 (floor)	2.0%	-11.3%	-0.65	-0.98	0.18	6.0%	65.69	30.1%	1.83	33.2%	6.73	-4.3%	0.0%
baseline, costs x2 (floor)	5.6%	-7.5%	-0.02	-0.04	0.75	9.3%	86.15	34.6%	1.85	52.9%	8.93	-0.6%	0.0%
strategy, costs x2 (real)	4.2%	-8.8%	-0.30	-0.46	0.48	6.0%	65.69	30.6%	2.00	33.2%	4.60	-2.2%	0.0%
baseline, costs x2 (real)	8.7%	-6.8%	0.28	0.43	1.28	9.3%	86.15	35.4%	1.98	52.9%	6.07	2.2%	0.0%
strategy, fill 1 bar later	1.3%	-6.9%	-0.80	-1.16	0.19	5.8%	88.31	29.3%	1.82	33.6%	4.57	-5.1%	0.0%
baseline, fill 1 bar later	9.3%	-6.3%	0.36	0.56	1.49	9.0%	86.15	35.4%	2.01	52.9%	4.48	2.7%	0.0%
strategy, best 10 trades removed	-3.1%	-13.0%	-1.14	-1.40	-0.24	7.8%	nan	na	nan	na	nan	na	na
param neighbour p=70, k=2.5	5.8%	-4.6%	-0.06	-0.10	1.26	5.1%	54.56	32.2%	2.05	27.1%	2.78	-0.3%	0.0%
param neighbour p=90, k=2.5	3.9%	-11.3%	-0.31	-0.48	0.35	6.6%	78.97	31.8%	1.90	39.3%	4.05	-2.5%	0.0%
param neighbour p=80, k=2.0	4.0%	-6.9%	-0.37	-0.58	0.57	5.6%	66.05	28.8%	2.16	30.2%	3.38	-2.1%	0.0%
param neighbour p=80, k=3.0	6.1%	-5.3%	-0.00	-0.01	1.15	6.0%	65.69	32.8%	2.00	35.2%	3.32	-0.4%	0.0%
param neighbour p=70, k=2.0	4.8%	-4.2%	-0.27	-0.43	1.13	4.8%	54.56	29.6%	2.14	24.6%	2.79	-1.3%	0.0%
param neighbour p=90, k=3.0	4.1%	-8.2%	-0.28	-0.43	0.50	6.7%	78.97	33.2%	1.81	41.4%	3.99	-2.3%	0.0%

Best 10 trades: Rs 4,664,786 of net trade P&L vs Rs -546,230 for all 183 test-window trades (positive net total = the best 10 are most of the profit; negative = they were offsetting losses).

Long and short legs separately (strategy is long-only by design; the symmetric-short version A1 failed validation): strategy test long-leg CAGR -1.0%, short-leg CAGR 0.0%; baseline long-leg 3.8%.

5. Stops, gaps and the per-trade loss cap

• Full period: 236 stop-outs out of 669 trades. Max loss at the stop price 0.75% of capital (cap 0.75%, hard limit 1%); max loss at the actual fill (gaps through the stop included) 1.89%.

• Test window: max loss at stop 0.74%; with gaps 1.89%; stop-outs with fill loss > 1% of capital (gap-driven): 5 (all windows).

• Train gap study: 39 of 1173 sessions (3.3%) opened more than 1% below the prior close (worst -9.2%); a gap of that size at full size costs more than 1% of capital. The overnight cap (8 lots) was tested and dropped on train (Sharpe collapsed because the overnight leg carries the return); the intraday-only variant D1 failed train (Sharpe -0.71).

6. Statistics

• 20-day block bootstrap (4000 draws), Sharpe(strategy) - Sharpe(baseline), test 2024-01..2026-10: median -0.63, 95% CI [-1.45, +0.04] (CI includes 0: not distinguishable)

• 20-day block bootstrap (4000 draws), Sharpe(strategy) - Sharpe(baseline), recent 2022-01..2026-10: median -0.42, 95% CI [-1.01, +0.13] (CI includes 0: not distinguishable)

• Configurations tried (unique idea x parameter cells, all phases): **127**. Deflated Sharpe Ratio on the test window: SR -0.09 annualised vs the expected best-of-127-noise SR0 1.81; DSR (probability the true Sharpe > 0 after deflation) = 0.00.

• Probability of Backtest Overfitting (CSCV, 10 partitions, 96 candidate configs, 2017-04..2023-12): **49%**.

7. Lookahead tests

• cut 2025-01-16: truncation (strategy signals) identical; truncation (baseline) identical; shuffled-future signals before cut identical; shuffled-future daily P&L before cut identical

• cut 2023-07-17: truncation (strategy signals) identical; truncation (baseline) identical; shuffled-future signals before cut identical; shuffled-future daily P&L before cut identical

• cut 2022-08-29: truncation (strategy signals) identical; truncation (baseline) identical; shuffled-future signals before cut identical; shuffled-future daily P&L before cut identical

• Extra delay (fill one bar later), test window: strategy Sharpe -0.09 -> -0.80; baseline 0.46 -> 0.36 (large collapse would mean the edge lives in the fill bar).

8. Synthetic series vs real NSE futures (daily bhavcopy): is the overnight edge real?

• train: mean night return real futures +2.7 bp vs synthetic +14.0 bp; mean day return real +2.0 bp vs synthetic -9.9 bp (n=1170)

• validate: mean night return real futures +1.3 bp vs synthetic +5.8 bp; mean day return real +1.5 bp vs synthetic -3.1 bp (n=467)

• test: mean night return real futures +0.7 bp vs synthetic +2.3 bp; mean day return real -2.2 bp vs synthetic -3.0 bp (n=682)

• Correlation night real vs synthetic 0.94, day 0.96. The synthetic series inherits the spot index's 09:15 "open"; if the night-leg means differ, overnight-heavy results (this strategy holds mostly overnight) are overstated on the synthetic series. Intraday real futures bars are not in the data, so this cannot be corrected here.

8b. Robustness: same signals on a REAL-futures-anchored price path (each session's open and close pinned to the actual NSE futures open/close; coverage 98% of sessions)

Signals are unchanged (spot). Only the P&L path changes, so overnight gaps equal real futures gaps. Not a gating test (the protocol's official yardstick is the synthetic series), but it shows how much of the result depends on the spot-open artefact.

	CAGR	MaxDD	Sharpe	Sortino	Calmar	Vol	TradesPerYr	WinRate	AvgWin/AvgLoss	TimeInMkt	CostDrag %/yr	LongLeg CAGR	ShortLeg CAGR
Train 2017-04..2021-12 — strategy	19.7%	-5.8%	1.65	2.90	3.39	7.4%	67.24	42.8%	2.23	40.0%	3.44	11.8%	0.0%
Train 2017-04..2021-12 — baseline	23.1%	-15.6%	1.11	1.77	1.48	14.0%	87.22	43.1%	1.97	56.7%	4.65	15.2%	0.0%
Train 2017-04..2021-12 — buy & hold futures	18.7%	-33.9%	0.70	0.99	0.55	18.0%	0.00	na	nan	99.9%	0.63	na	na
Validate 2022-01..2023-12 — strategy	12.6%	-3.9%	0.81	1.27	3.22	7.4%	88.61	37.6%	2.20	41.0%	4.46	6.9%	0.0%
Validate 2022-01..2023-12 — baseline	16.5%	-5.7%	1.01	1.58	2.88	9.5%	100.39	43.4%	1.76	54.0%	5.21	10.5%	0.0%
Validate 2022-01..2023-12 — buy & hold futures	13.0%	-15.3%	0.51	0.71	0.85	13.8%	0.00	na	nan	100.0%	0.61	na	na
TEST 2024-01..2026-10 (locked, opened once) — strategy	4.8%	-7.5%	-0.22	-0.34	0.63	5.8%	65.69	36.6%	1.56	32.8%	3.36	-1.7%	0.0%
TEST 2024-01..2026-10 (locked, opened once) — baseline	10.6%	-6.4%	0.47	0.71	1.67	9.5%	86.15	42.5%	1.57	52.9%	4.47	4.0%	0.0%
TEST 2024-01..2026-10 (locked, opened once) — buy & hold futures	1.0%	-15.8%	-0.33	-0.45	0.06	12.9%	0.00	na	nan	99.7%	0.61	na	na
Recent 2022-01..2026-10 — strategy	7.9%	-7.5%	0.26	0.41	1.05	6.5%	75.14	37.1%	1.86	36.2%	3.82	1.8%	0.0%
Recent 2022-01..2026-10 — baseline	13.0%	-6.4%	0.69	1.07	2.04	9.5%	92.02	42.9%	1.66	53.4%	4.77	6.7%	0.0%
Recent 2022-01..2026-10 — buy & hold futures	5.8%	-15.8%	0.03	0.04	0.37	13.3%	0.00	na	nan	99.8%	0.61	na	na
Full 2017-04..2026-10 — strategy	13.6%	-7.5%	0.99	1.65	1.80	7.0%	71.22	39.8%	2.07	38.1%	3.63	6.8%	0.0%
Full 2017-04..2026-10 — baseline	17.9%	-15.6%	0.92	1.45	1.15	12.0%	89.64	43.0%	1.82	55.0%	4.71	10.9%	0.0%
Full 2017-04..2026-10 — buy & hold futures	12.0%	-33.9%	0.41	0.57	0.35	15.8%	0.00	na	nan	99.9%	0.62	na	na

Comparison with simply holding Nifty futures (1x, always long, rolled; same costs, same interest convention)

	CAGR	MaxDD	Sharpe	Sortino	Calmar	Vol	TradesPerYr	WinRate	AvgWin/AvgLoss	TimeInMkt	CostDrag %/yr	LongLeg CAGR	ShortLeg CAGR
Train 2017-04..2021-12 — buy & hold (synthetic)	17.1%	-33.7%	0.63	0.88	0.51	17.7%	0.00	na	nan	99.9%	0.63	na	na
Train 2017-04..2021-12 — strategy (synthetic)	19.3%	-6.2%	1.68	2.95	3.11	7.1%	67.24	43.1%	2.19	40.7%	3.44	11.4%	0.0%
Validate 2022-01..2023-12 — buy & hold (synthetic)	12.8%	-16.0%	0.50	0.69	0.80	13.8%	0.00	na	nan	100.0%	0.61	na	na
Validate 2022-01..2023-12 — strategy (synthetic)	10.7%	-5.5%	0.60	0.93	1.94	7.2%	88.61	37.6%	2.06	41.8%	4.44	5.2%	0.0%
TEST 2024-01..2026-10 (locked, opened once) — buy & hold (synthetic)	2.8%	-14.6%	-0.20	-0.28	0.19	12.7%	0.00	na	nan	99.7%	0.61	na	na
TEST 2024-01..2026-10 (locked, opened once) — strategy (synthetic)	5.5%	-8.7%	-0.09	-0.15	0.64	5.9%	65.69	31.1%	2.09	33.2%	3.36	-1.0%	0.0%
Recent 2022-01..2026-10 — buy & hold (synthetic)	6.8%	-16.0%	0.10	0.14	0.42	13.1%	0.00	na	nan	99.8%	0.61	na	na
Recent 2022-01..2026-10 — strategy (synthetic)	7.6%	-8.7%	0.23	0.35	0.88	6.5%	75.14	34.3%	2.07	36.7%	3.81	1.6%	0.0%

	CAGR	MaxDD	Sharpe	Sortino	Calmar	Vol	TradesPerYr	WinRate	AvgWin/AvgLoss	TimeInMkt	CostDrag %/yr	LongLeg CAGR	ShortLeg CAGR
Full 2017-04..2026-10 — buy & hold (synthetic)	11.8%	-33.7%	0.40	0.55	0.35	15.6%	0.00	na	nan	99.9%	0.62	na	na
Full 2017-04..2026-10 — strategy (synthetic)	13.3%	-8.7%	0.98	1.62	1.53	6.8%	71.22	38.4%	2.17	38.7%	3.63	6.5%	0.0%

Regime-wise results inside train / validate / test (annualised return contribution and Sharpe of futures P&L after floor costs; regimes from prior-day data)

	days	ann_return	ann_sharpe
Train 2017-04..2021-12 — all days — strategy	1173	11.8%	1.68
Train 2017-04..2021-12 — all days — baseline	1173	15.3%	1.15
Train 2017-04..2021-12 — all days — buy&hold	1173	11.2%	0.63
Train 2017-04..2021-12 — uptrend (>200DMA) — strategy	763	8.2%	1.70
Train 2017-04..2021-12 — uptrend (>200DMA) — baseline	763	9.3%	1.51
Train 2017-04..2021-12 — uptrend (>200DMA) — buy&hold	763	4.3%	0.48
Train 2017-04..2021-12 — downtrend (<200DMA) — strategy	410	3.6%	1.64
Train 2017-04..2021-12 — downtrend (<200DMA) — baseline	410	6.1%	0.94
Train 2017-04..2021-12 — downtrend (<200DMA) — buy&hold	410	6.9%	0.85
Train 2017-04..2021-12 — high VIX (pct>80) — strategy	285	0.0%	0.32
Train 2017-04..2021-12 — high VIX (pct>80) — baseline	285	1.4%	0.28
Train 2017-04..2021-12 — high VIX (pct>80) — buy&hold	285	-3.9%	-0.61
Train 2017-04..2021-12 — normal VIX — strategy	888	11.8%	1.93
Train 2017-04..2021-12 — normal VIX — baseline	888	13.9%	1.93
Train 2017-04..2021-12 — normal VIX — buy&hold	888	15.2%	1.46
Validate 2022-01..2023-12 — all days — strategy	492	4.3%	0.60
Validate 2022-01..2023-12 — all days — baseline	492	6.8%	0.73
Validate 2022-01..2023-12 — all days — buy&hold	492	6.8%	0.50

	days	ann_return	ann_sharpe
Validate 2022-01..2023-12 — uptrend (>200DMA) — strategy	386	2.2%	0.40
Validate 2022-01..2023-12 — uptrend (>200DMA) — baseline	386	3.4%	0.55
Validate 2022-01..2023-12 — uptrend (>200DMA) — buy&hold	386	-1.5%	-0.16
Validate 2022-01..2023-12 — downtrend (<200DMA) — strategy	106	2.1%	1.26
Validate 2022-01..2023-12 — downtrend (<200DMA) — baseline	106	3.4%	1.21
Validate 2022-01..2023-12 — downtrend (<200DMA) — buy&hold	106	8.4%	2.17
Validate 2022-01..2023-12 — high VIX (pct>80) — strategy	62	0.5%	2.64
Validate 2022-01..2023-12 — high VIX (pct>80) — baseline	62	2.8%	1.67
Validate 2022-01..2023-12 — high VIX (pct>80) — buy&hold	62	4.6%	1.68
Validate 2022-01..2023-12 — normal VIX — strategy	430	3.8%	0.57
Validate 2022-01..2023-12 — normal VIX — baseline	430	3.9%	0.53
Validate 2022-01..2023-12 — normal VIX — buy&hold	430	2.3%	0.21
TEST 2024-01..2026-10 (locked, opened once) — all days — strategy	702	-0.6%	-0.09
TEST 2024-01..2026-10 (locked, opened once) — all days — baseline	702	4.2%	0.46
TEST 2024-01..2026-10 (locked, opened once) — all days — buy&hold	702	-2.6%	-0.20
TEST 2024-01..2026-10 (locked, opened once) — uptrend (>200DMA) — strategy	452	-2.2%	-0.67
TEST 2024-01..2026-10 (locked, opened once) — uptrend (>200DMA) — baseline	452	-0.3%	-0.05
TEST 2024-01..2026-10 (locked, opened once) — uptrend (>200DMA) — buy&hold	452	-3.2%	-0.44
TEST 2024-01..2026-10 (locked, opened once) — downtrend (<200DMA) — strategy	250	1.6%	0.65
TEST 2024-01..2026-10 (locked, opened once) — downtrend (<200DMA) — baseline	250	4.5%	1.16
TEST 2024-01..2026-10 (locked, opened once) — downtrend (<200DMA) — buy&hold	250	0.6%	0.12

	days	ann_return	ann_sharpe
TEST 2024-01..2026-10 (locked, opened once) — high VIX (pct>80) — strategy	180	-0.6%	-0.88
TEST 2024-01..2026-10 (locked, opened once) — high VIX (pct>80) — baseline	180	3.7%	1.07
TEST 2024-01..2026-10 (locked, opened once) — high VIX (pct>80) — buy&hold	180	5.7%	1.29
TEST 2024-01..2026-10 (locked, opened once) — normal VIX — strategy	522	-0.0%	-0.00
TEST 2024-01..2026-10 (locked, opened once) — normal VIX — baseline	522	0.5%	0.10
TEST 2024-01..2026-10 (locked, opened once) — normal VIX — buy&hold	522	-8.2%	-1.04

- H1 (OI x price, prior-day data, n=1) on train, all days: Sharpe 0.40, MaxDD -18.3%, CAGR 12.3%
- H1 (OI x price, prior-day data, n=1) on train, excluding sessions within 3 days of expiry: Sharpe 0.25, MaxDD -19.4%, CAGR 9.3%
- OI lookahead: OI is published after the close, so day d uses only day d-1 OI (signal shifted one day in `Sig.H1`); H1 is not part of the frozen strategy.

9. Pass criteria (all must hold) and roadmap targets

Criterion	Result	Detail
Sharpe above baseline on test (floor)	FAIL	-0.09 vs 0.46
Sharpe above baseline on 2022-01..2026-10 (floor)	FAIL	0.23 vs 0.57
Sharpe above baseline on test (realistic costs)	FAIL	0.09 vs 0.62
MaxDD better than -15% on every window and better than the baseline's	FAIL	train: -6.2% vs -13.4%; valid: -5.5% vs -7.7%; test: -8.7% vs -6.6%; recent: -8.7% vs -7.7%; full: -8.7% vs -13.4%
Walk-forward Sharpe above baseline's, drawdown better than -15%	FAIL	Sharpe 1.12 vs 1.29; MaxDD -6.2% (baseline -7.7%)
No calendar year worse than -10%	PASS	worst -3.3% (2024)
Still beats the baseline with doubled costs (test, floor and real)	FAIL	floor x2: -0.65 vs -0.02; real x2: -0.30 vs 0.28
No trade lost more than 1% of capital at its stop (gaps reported separately)	PASS	max 0.75% at the stop; 1.89% incl. gaps

Roadmap targets (floor costs):

Target	Met?	Value
Sharpe Apr 2017-Oct 2026 >= 1.0	no	0.98 (baseline 0.90)
Sharpe Jan 2022-Oct 2026 >= 0.8	no	0.23 (baseline 0.57)
Max drawdown better than -12% (hard limit -15%)	yes	-8.7% (baseline -13.4%)
Walk-forward CAGR >= 15% and MaxDD better than -15%	no	CAGR 14.5%, MaxDD -6.2% (baseline 20.8% / -7.7%)

Target	Met?	Value
Sharpe on locked test above baseline	no	-0.09 vs 0.46
Loss per trade at the stop <= 0.75% (cap 1%)	yes	0.75%

10. Gate G5 verdict: NO-GO: not all pass criteria hold

On the real-anchored path (section 8b) the test-window Sharpe is -0.22 for the strategy vs 0.47 for the baseline and -0.33 for buy-and-hold futures; full-period 0.99 / 0.92 / 0.41.

Honest limits: (i) Gate G3 not met (only one idea validated); (ii) prices are synthetic futures from spot, and section 8 quantifies how the overnight leg differs from real futures; (iii) the test window's Sharpe has a standard error near 0.6, so no single number proves superiority; (iv) earlier work in this repo had already looked at 2024-26 for other strategies; (v) two implementation corrections were made after the first validation looks (PROTOCOL_CHANGES.md).